

Impact Statement

This project fine-tunes a transformer to classify consumer support complaints into ten categories (e.g., billing, fraud_unauthorized, account_access), aiming to speed up and standardize how complaints are routed to staff.

Positive impacts.

If successful, the model reduces the manual burden on support staff who currently sort large volumes of ambiguous text by hand, freeing time for actual resolution rather than triage. Consumers benefit from faster, more consistent routing, avoiding the common frustration of repeating their issue to multiple departments. Most importantly, accurate classification could help time-sensitive complaints, such as fraud or account access, reach the right team faster than they would in a manual queue, which matters when delay itself causes harm (e.g., an unresolved account breach).

Risks.

The training set is small (380 labeled examples across 10 categories) and its demographic and geographic provenance is undisclosed. This raises two concrete risks. First, rarer categories are likely underrepresented, so the model may perform worst exactly where mistakes carry the highest stakes: misrouting a fraud complaint as "general_inquiry" could delay a consumer's ability to protect their finances. Second, because we cannot verify who the data represents, the model's behavior for underrepresented populations or complaint-writing styles is unknown, and could unintentionally systematize unequal service quality. Over time, if such a system were deployed at scale without human oversight, misclassifications could compound: consumers with less "typical" phrasing might be routed incorrectly more often, eroding trust and autonomy for those already less well served by existing support systems.

Mitigations and limitations.

We do not treat automated classification as a replacement for human judgment in sensitive cases; our stated design value is that humans remain in the loop for ambiguous or high-stakes complaints (e.g., fraud, account access) rather than allowing the model to route or resolve these independently. We are also considering data augmentation to address class imbalance, though this cannot fully substitute for more representative source data. We are explicit that this is a small-scale academic prototype, not a validated production system, and we make no claims about its reliability outside this evaluation setting.

Uncertainty.

We do not know how the model will behave on complaint phrasing or demographics absent from the 380-example training set, and we have not conducted a formal fairness audit across categories. We view transparency about these gaps, rather than overstating readiness, as itself part of the project's commitment to trust and accountability.